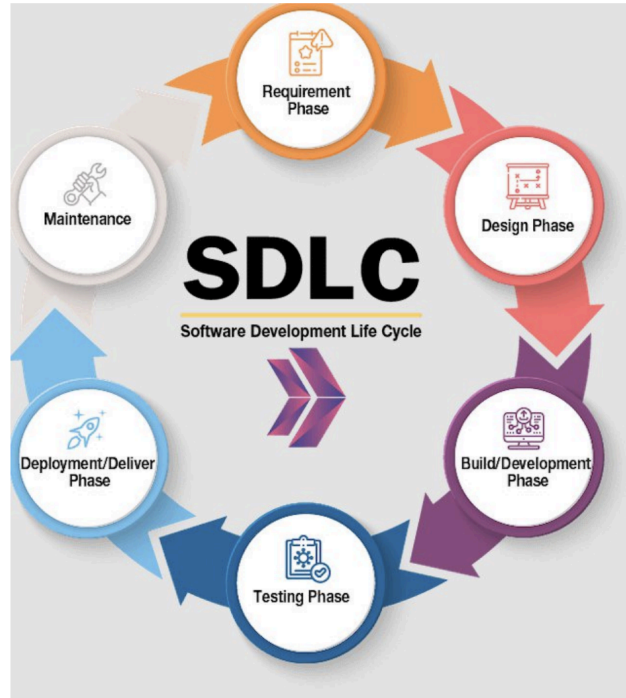


Q1

Software Development Life Cycle (SDLC)



Requirement Analysis: The teams captures customers requirements and develops functional requirements which are then documented in the Business Requirement Document (BRD) and Software Requirement Specification (SRS).

Design: System architects create the High-Level Design (HLD) for overall architecture and Low-Level Design (LLD) for module-specific details, APIs, and database

Development phase: Programmers code source code, code review, do unit testing, and version control according to the design that is accepted.

Testing: QA engineers design and execute test cases (manual, automated, regression, security) to find and fix defects before release.

Deployment: Once verified, the software is published to the production server where end-users can access it.

Maintenance: Ongoing bug fixing, better performance, and security updates are done to keep the application running over time.

Q2

- User Story 1 (Product Search & Filtering): As a shopper, I want to filter products by category, price, and brand so that I can quickly find the exact items I want to buy.
- User Story 2 (Shopping Cart): As a shopper, I want to adjust item quantities or remove items directly in my shopping cart so that I can manage my total order cost before checkout.
- User Story 3 (User Reviews): As a shopper, I want to leave a rating and photo review on products I purchased so that I can share my feedback with other buyers.
- User Story 4 (Seamless Checkout): As a returning customer, I want to save my shipping addresses and payment options so that I can complete my purchases quickly.
- User Story 5 (Order Tracking): As a buyer, I want to view real-time status updates and shipping tracking links for my past orders so that I know when to expect delivery.

User Story 2 (Shopping Cart)

- **Acceptance Criterion 1:** Given a user is on the cart page, when they click the + or - buttons next to an item, the quantity updates immediately without requiring a full page refresh.
- **Acceptance Criterion 2:** Given the item quantity changes, the total price for that item line and the overall cart total must update automatically in real time.
- **Acceptance Criterion 3:** Given an item quantity is set to 0 or the "Remove" icon is clicked, the system prompts a brief confirmation before removing the item from the cart.

- **Acceptance Criterion 4:** Given a product in the cart goes out of stock while in the user's session, an warning badge must display stating "Out of Stock," disabling the checkout button until resolved.
- **Acceptance Criterion 5:** Given a user is not logged in, cart contents must persist across session refreshes using browser local storage for up to 7 days.

Task ID	Item / User Story	Technical Sub-Tasks	Estimated Effort	Priority
TSK-01	UI Component Setup	Design cart UI components (quantity toggles, remove buttons, total summary box).	3 Story Points	High
TSK-02	Real-time Cart State	Implement frontend state management for dynamic price updates.	5 Story Points	High
TSK-03	Inventory API Sync	Build REST API endpoint to validate live product availability .	5 Story Points	High
TSK-04	Session & Account Merge	Store guest cart in local storage and merge items upon user login	3 Story Points	Medium
TSK-05	Edge Case Handling	Implement low-stock alerts, out-of-stock badges, and confirmation.	2 Story Points	Medium

What the AI did well:

It quickly structured the user stories using the standard format. Ai saved time with the workflow and acceptance criteria in seconds.

What corrections did i make

Formating fix i change manually .The acceptance criteria to handle inventory shortages nicely and also added account cart-merge

Why human review is still necessary:

AI gives common templates, but human review is required to catch missing business logic that ai have missed and user experience details that only human can understand and make a logic out of it.And the testing scenarios that the AI overlooked and might think that it not really important

Ostad - Learn Skills Live

AI Software Planning Exercise

+

gemini.google.com/app/9091e262e51625ee

Ask Gemini

explain the q2 i cant undrestand what is i have to do

Upgrade

Question 2 is a hands-on exercise designed to test how you use AI to speed up software planning and where human supervision is required. Here is a clear, step-by-step breakdown of what you need to do:

Step-by-Step Action Plan

- **Pick an AI Tool:** Choose one AI assistant (ChatGPT, Gemini, Claude, GitHub Copilot, or Cursor AI).
- **Prompt the AI for User Stories:** Ask your chosen AI to write **5 user stories** for an e-commerce website (e.g., shopping cart, checkout, product search, user reviews, or wishlists).
- **Generate Acceptance Criteria:** Ask the AI to pick **one** of those user stories and break it down into at least **5 acceptance criteria** (the conditions that must be met for the feature to be considered complete).
- **Create a Sprint Backlog:** Ask the AI to turn those stories/criteria into a **Sprint Backlog** (a prioritized list of tasks for a development sprint).
- **Review & Refine (Human Touch):** Read through the output. Fix errors, add missing edge cases, or adjust unclear logic (e.g., adding security rules to checkout or clarifying payment failure steps that the AI missed).

+ Ask Gemini

Flash

Gemini is AI and can make mistakes.

The screenshot shows the Google Gemini web interface. At the top, there's a browser tab with the URL 'gemini.google.com/app'. The main content area displays a conversation with Gemini. A system message (blue bubble) says: 'now take the action plan step by step and do the work what has been told from first to Ask the AI to create a Sprint Backlog.' Below this, a user message (pink bubble) asks: 'Here is the complete initial workflow from generating the user stories down to the Sprint Backlog.' Gemini's response (blue bubble) is: '1. Generate 5 User Stories for an E-Commerce Application' followed by a list of five user stories. At the bottom, there's a search bar with the text 'Ask Gemini' and a 'Flash' button. A small disclaimer at the very bottom reads: 'Gemini is AI and can make mistakes.'

now take the action plan step by step and do the work what has been told from first to Ask the AI to create a Sprint Backlog.

Here is the complete initial workflow from generating the user stories down to the Sprint Backlog.

1. Generate 5 User Stories for an E-Commerce Application

- **User Story 1 (Product Search & Filtering):** As a shopper, I want to filter products by category, price, and brand so that I can quickly find the exact items I want to buy.
- **User Story 2 (Shopping Cart):** As a shopper, I want to adjust item quantities or remove items directly in my shopping cart so that I can manage my total order cost before checkout.
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Ask Gemini

Gemini is AI and can make mistakes.